

Questions

1. Where is the fuel being stored in the body
2. How does data transmission b/w RPA's and controller takes place
3. ~~How does~~ What kind of logs are being stored and retrieved from Helicopter
4. What is drone train?
5. Why there isn't a EV Helicopter yet?

Day 4 :RPA S

main rotor

- 200 kg weight, 2 bladed main & tail rotors
- Transportation of cargo to & from high altitude (forward bases of armed forces?)
- material used :

Combination of metallic materials for better strength to weight ratio and stiffness to weight ratio

- Composite materials - resistance to corrosion, reduce maintenance requirements, direct opening cost

- mostly Al alloy

- Configuration :

no of blades - 2 :

diameter - 4.6 m

840 RPM

ang velo : 87.96 rad/sec

main rotor hinge less

chord - 0.17

tip speed - 200 m/s

twist - 12°

no taper

main rotor

Tail rotor config

type : teetering type

side type : pusher

no of blades : 2

4010 RPM

ang vel : 420 rad/sec

structure :

front fuselage center fuselage Tail boom

Avionics :

flight control unit

Data link

Propulsion :

- Engine
- fuel system

Communication : ground control system

Power source

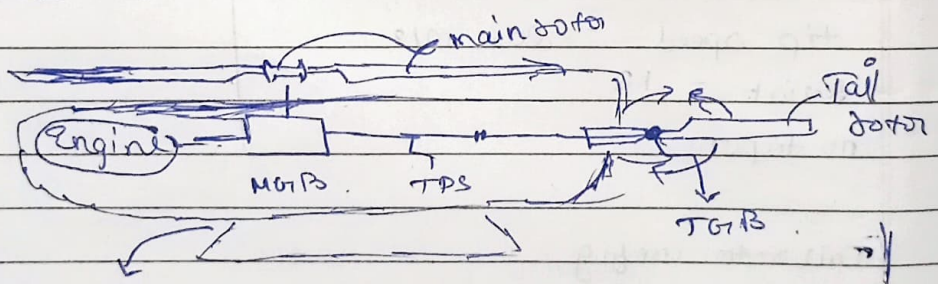
- on board ~~power~~ elec power system
- Transmission system

Payload : EO system

main rotor

- 200 kg
- rotor blades attached to blade grip with 2 bolts then mounted on metallic bearings in blue bearing sleeve & cross bar module
- bearing enable feathering action of the blades and hold blades against centrifugal force

drive Train :



landing gear

- skid landing gear
- absorbs landing impact energy by deflection of cross tubes
- permanent deformation of cross tube is permitted during landing at 2 m/s sink velocity

Transmission System

- power from engine to a rotor rotor
- This includes main gear box (MOB), tail gear box (TOB), tail drive shaft (TDS); flexible drive shaft (FDS), free wheel (FW) & gearbox

main gear box

- Engine output flange - 87150 rpm - connected to MOB input flange through main drive flexible shaft
- Employs 2 stage speed reduction gear box
- main rotor - 838.35 rpm
- Tail rotor - 4014.35 rpm

Fuel System

- Air frame fuel system, 55 kg fuel capacity, with filler cap for gaseous fueling

operating fuel : Avgas / gasoline

fuel tank volume : 55 kg

flow rate : 60 lph

fuel delivery pressure : 1 to 2 bar

Anti Collision light

- installed in vertical fin of RPA to avoid collision with other flying objects including air crafts, helicopters & other UAVs
- it is Not friendly, aviation red color



classmate

Date _____
Page _____

Visual Learning

BP viewing

Windchill, Teamcenter
↳ part no indexing

software used

So cool!

APRIS Commander
desktop CAD

Geo view
lite

Simulation

Simons

ADERT System

SOP - standard of preparation

IPC - Illustrated parts Catalogue

user manual created using

~~Robot Systems~~

Each parts has its Indexing

- manual for understanding & problem solving & knowing the procedure
- for assembling and disassembling

room diagram - ~~the~~ electrical diagram

Component, illustration, part number

design → production team - testing

↓
used IPC & user manual for
assembling / gathering /
manufacturing parts

Autopilot system:

Fight Control Computer interfaced with
(AIRS, ADU, laser altimeter) as basic
flight control & navigation sensors

- Capable of driving electromechanical
servo actuator
- has compatible ground control system,
controlling the RPAS payloads
- Command - Control data link system b/w
RPAS & GCS.

Operations

FCU Enables

- fully autonomous mode - manual mode
- both combination
- can switch b/w these modes through commands from GCS: manual to automatic
- control system on FCU provides stability & augmentation

Problems

- vibration fault / temp / over speed from engine
- Human errors on tap testing of wings - CV

FCU provides

- air speed hold, ground speed hold mode,
- Baro altitude hold mode
- AGL Height hold mode during hover and cruise
- automatic waypoint navigation
- Automatic take off & landing (ATOL)

Control System - Safety modesFCU?

- Return home in case of link loss
- autonomous autorotation followed by flare
- touch down in case of loss of engine power or loss of tail rotor thrust
- automatic landing function after powered descent in case of multiple in flight failure
- operates at multiple speed & altitude
- controls (EO / IR cameras) payloads integrated with RPA's

Electro optical system

Data link System:

- down linking of payload video and telemetry data from the aircraft to ground control station (GCS)
- up linking of command & control data for payload from the ~~GCS~~ GCS

a) ~~ADT~~ Airborne data link terminal (ADT)

b) ground data link terminal (GDT)

ADT:

- radio, omni directional antenna, GPS antenna

GDT:

a) GDT radio

b) ground omni directional antenna

c) switch beam antenna

d) GPS antenna

e) directional antenna

f) Antenna Position Unit

g) pendant

- GDT LRU are mounted on RPAS ground control system?
- ADT and GDT are interfaced to host system via Ethernet & ^{serial} ~~serial~~ interfaces

a) operating modes following are data link system operating modes

- direct LOS mode (ADT-GDT point to point operation)

b) NLOS mode?

data link system : band - 4.4 GHz to 5.6 GHz
uplink 9.6 kbps
downlink video data rate : 4 Mbps

GDT receives radio signal using switch beam antenna (SBA) and directional antenna
- SBA can electronically steer its beam without physically moving the antenna

Lidar : Sense distance b/w UAV & ground, with max range of 100 m
- helpful in take off & landing for find the vertical speed w.r.t to ground distance
Laser

Pitot static probe - air speed measurement

Questions to Ask :

1. what takes people the most time?
2. what paperwork do they still have?
3. data for that exist but nobody uses it?
4. what decisions are repetitive?
5. what inspections are manual?
6. what reports are manually generated?
7. what causes delays?
8. what mistakes happen frequently?

ALH

Day 5

DHRUV - MK III - Flight manual

main rotor diameter : 13.2 m

fuel : 1352 l (1055 kg)

anti icing additives below 15°C flight above 3000 m, not permitted without
fitment & oxygen for crew

- take off and landing on ground above 10° is not advised
- spot turns (turns whilst hovering)
with greater than $60^{\circ}/\text{sec}$

Graphs

$\uparrow$ load factor vs $\downarrow$ ~~load~~ speed
 $\downarrow$ NZ vs IAS (kmph)
 (load factor)

vibration monitoring

Maintenance manual

VMS - vibration monitoring system

00, 02, 03

- NBS software is used to retrieve the data and in the form of .obs

- AHRS - attitude heading reference system

- attitudes • attitude rates • yaw rates, heading & accelerations to AFCS and to IADS

- IADS - integrated architecture display system

Vibration mode Exceedence limit are different for ALH, LCH

- Extraction of VMS data takes place using cockpit control head (CCH)

warnings

- 2 monitor modes, measurement mode to measure, record & trigger warnings vibrations above threshold:
- first monitor continuously monitor tail rotor 1/rev unbalance in axial & radial direction
- Second monitor - main rotor 1/rev - continuously - unbalance in lateral and vertical direction
- • Engine overall vibration every 5 min

Threshold for tail rotor vibration - High & Green during monitor mode

	Axial		Radial	
	High	Green	High	Green
Tail rotor	1.0 ips	1.5 ips	1.5 ips	2.5 ips

Threshold of main rotor vib during measurement mode

	Axial		Radial	
	High	Green	High	Green
Tail rotor	0.7 ips	1.0 ips	0.7 ips	1.5 ips

main rotor	0.5 ips	0.7 ips	0.5	0.7
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Data downloading using PC-XTBS



IPS - inch per sec

Portable computer

ground based software

Day 6

- Study about materials which can have minimal effect on aircraft during a crash / inadvertent event of a Helicopter

Properties:

- Strength, Stiffness, Density, percentage elongation, Good fatigue strength, behavior at elevated temperature, Resistance to corrosion, Ease of fabrication, cost

materials under study:

- Al alloy, Titanium alloy, Alloy steel, Mg Alloy, Copper alloys, Carbon prepreg, Kevlar prepreg, perspex, Rubber

Structural build:

- Monocoque, - Semi monocoque, - Sandwich, - sheet metal, - machined, - composite

different construction of tail boom:

- Torus, Semi monocoque, Sandwich

Mechanism of crash

- ° Crash - sudden reduction of velocity - deceleration - measured in g
- ° higher g - worst the situation

- To achieve: design a select materials in such a way to have minimal impact to the crew seat & bottom structure

Fuel cells - cause fire

$$\text{margin of safety } MS = \frac{F_{ult}}{F_{den} \times 1.5} - 1$$

- Test of Emergency floating gear (EFG)
- hard Impact test
- Lightning protection tests for direct effects

Helicopters: Dthaw - skid & wheel

RWRDC - Rotary wing research & design center

MRO - Maintenance, Repair & overhaul

- foldable tail boom

Day - 8

projects on mind

1. Creating a database of emergency landing for a particular region from satellite images
2. Study how vibration data can be used, what can be built - prediction model / detection model?
3. Advanced CNN data preprocessing
4. automated search & rescue using CV

Questions I must get an answer to

1. What makes an ~~helicopter~~ landing site safe for an helicopter?
2. This helps me define features, create a scoring system, and train a model

Task with Test pilots / flight operations / flight safety

1. what factors determine whether a location is suitable for emergency landing
2. what is the first thing pilot looks for from the air
3. what type of terrain is rejected
4. how much flat area is required
5. how much slope is acceptable
6. how close can obstacles be
7. is wind acceptable?
8. what is the difference b/w normal & emergency landing decisions

Certification departments

1. are there landing site requirements?
2. what defines a safe landing zones

Operations Engineers

1. How are Emergency landing sites are identified today?
2. is there any software doing this
3. what info is useful during an Emergency
4. What is current difficulty

Manuals to read

1. flight manuals.
 - ↳ helicopter limitations
 - operating conditions
 - Emergency protocols

helicopter
performance

2. System description manuals

Day - 9

we are studying about LCH (Light Combat Helicopter)

General:

- Twin Engine

Features:

- Reduced visual, aural, radar & IR signatures, crashworthy landing gear, better survivability

Airframe

- Cockpit & pilots
- Tailboom with empennage

Landing gear

- non retractable tail wheel tricycle type with crashworthy features
- fitted with oleo-pneumatic shock absorbers

Transmission

- Both engine powers $\rightarrow$ main gear box $\rightarrow$ main rotor
- ~~MCB~~ MCB $\rightarrow$ Tail drive shaft $\rightarrow$ tail rotor by flexible coupled shafts
- TDS supports break assembly, intermediate gear box, "Tail gear box"

Fuel system

- Suction fed system $\rightarrow$ LH & RH engine fed lines
- 3 main tanks & 2 supply tanks
- main tanks inter connected to each other

Rotor:

- main: 4 composite hingeless blades
- main rotor hub & upper control system position inside gearbox housing & rotor drive shaft
- combination of MCB, upper control & rotor head - Integrated dynamic system (IDS)

Hydraulic System

- 2 Independent, provide hydraulic power simultaneously to both main & tail rotor flight control actuators.

Flight control:

- hydraulic powered mechanical system
- main rotor system $\left\{ \begin{array}{l} \text{upper control} \rightarrow \text{integrated in rods} \\ \text{lower control} \end{array} \right.$
- lower control of main rotor - collective control, cyclic longitudinal & cyclic lateral control
- upper control system - mixing unit assembly, swash plate mast, swash plate, rotating control rods, non rotating control rods, rotating & non rotating pinions
- adjustable / conventional directional control pedals are provided in the cockpit for tail rotor control

Automatic flight control system (AFCS):

- handling qualities, trim-facilities, autopilot modes
- dual, microprocessor based dual systems

Electrical System

- have both AC & DC generating systems
- DC: 2 independent subsystem - generator, a battery, a control & protection system
- both DC faults - Emergency DC loads supplied to battery
- AC: 2 independent subsystem, each has alternator - together associated with control & protection system
- both faults, Emergency AC loads supplied from a static inverter

Important data :

Seating : 2 pilots + 1030

maximum take off weight - 5200 kg

Engine : Shakti - 1411

output shaft speed - 6000 RPM

take off power - 1032 kW

Main gear box :

Input speed (Engine - HUB) - 6000 RPM

main rotor drive speed - 314 RPM

Intermediate gear box :

IP speed - 4163 RPM

OP speed - 4033 RPM

Tail gear box :

IP speed - 4033 RPM

Tail rotor drive speed - 1564 RPM

Oil cooling Sys :

Engine : Integrated oil cooling System

HUB : Separate system with own cooler & fan

Fuel System :

Fuel System : ATF K-50

Total capacity : 1032 kg



Main rotor

no of blades - 4

Rotor diameter - 13.2 m

direction of rotation : clockwise from top

Rotational speed - 314 RPM

Tail Rotor :

no of blades - 4

Rotor diameter : 8.55 m

direction of rotation : bottom moves forward

Rotation speed : 1564 RPM

(ATA-100 standards)

- 17 technical manuals for LCH
- Data module code - standardized & structured identifier of data module

Data module code

M1 - SDC - SNS - (DC) (DCV) - (IC) (ICV) - (LIC)

M1 - Model Identification

SDC - System difference code

{	AO - army
	BO - airforce
	A3 - both

SNS - standard numbering system

DC : disassembly code

DCV - disassembly code variant

IC - information code

ICV - info ... code variant

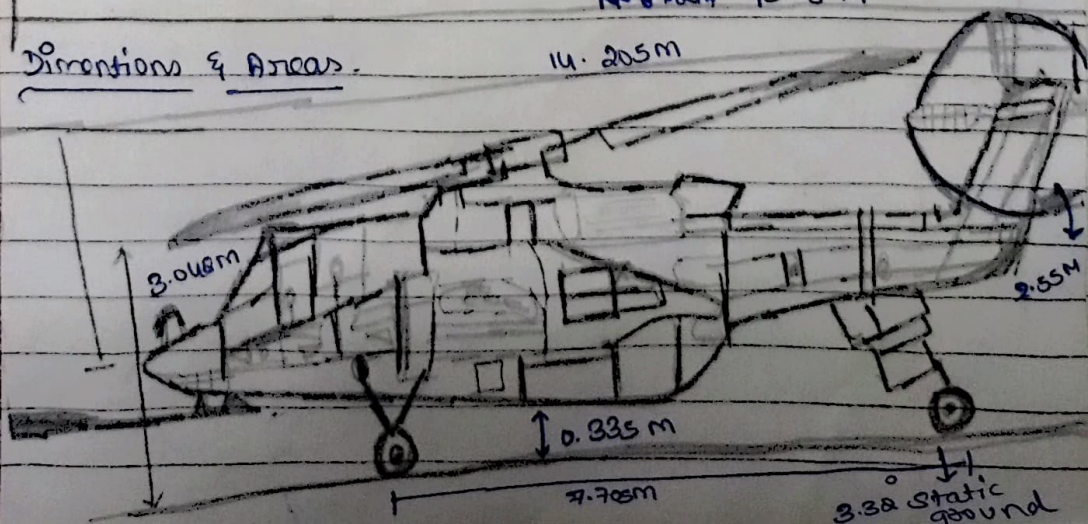
LIC - item location code

A - item installed on helicopter

B - items installed on major assembly removed from

C - items on bench (removed from helicopter) helicopter

D - info related to all 3 locations A, B & C

~~15.8 M~~ 15.8 MDimensions & Areas

Total length: 15.8 m

From cockpit nose:

tail boom end: 14.205 m

Rotor wing end: 14.876 m

14.205

14.876

(main) Rotor wing to ground: 3.048 m

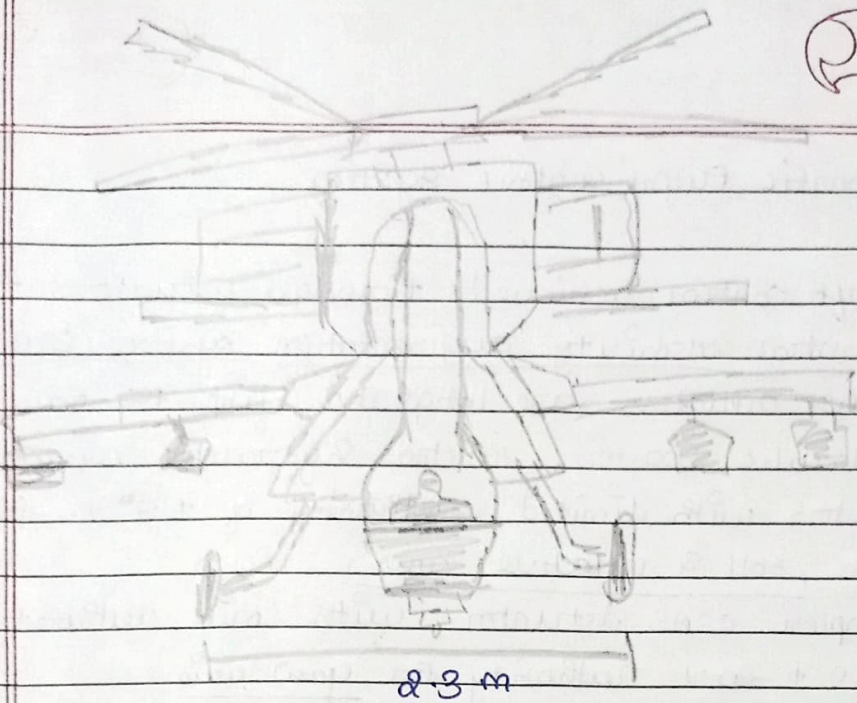
tail rotor top wing's top to ground: 4.36 m

3.048
m

4.36m

13.8 m

1.999 m



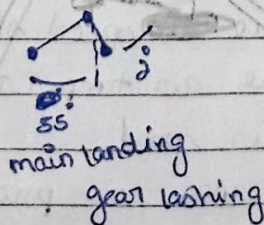
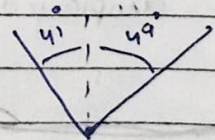
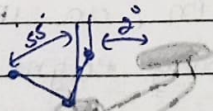
2.3 m

5.15 m

Lashing: secure/immobilized the helicopter when parked

- 4 lashing rings are provided
- operation to secure the helicopter which has just landed

- The helicopter is positioned with its nose tip facing the wind
- pickets / anchoring points are driven into the ground
- Ropes are tied b/w lashing rings of helicopter & anchoring points on ground.



Tail landing gear
lashing

Automatic Flight Control System

aircraft control achieved through actuators - control stability augmentation system (CSAS)

- CSAS actuators are integrated with the main hydraulic boosters, electro hydraulic servo actuators with limited authority of $\pm 10\%$ for pitch, roll & collective axis
- Simplex CSAS actuator with min authority of $\pm 20\%$ authority for yaw axis
- Auto trim achieved through parallel actuators in pitch, roll, yaw and collective with 100% authority
- Trim / force feel system - electro mechanical system ~~which~~ - operated directly through manual control, as well as by AFCS
- Trim / FFS - spring based unit for pitch, roll axes, friction based force feel system in yaw and collective axes - yaw / collective axes
- Has in flight monitoring system - detect failure during flight
- If any malfunction, on IADS, also even displayed on primary flight display (PFD)

ADU:

- PFD provides baro altitude, indicated airspeed and true airspeed signals to AFCS
- measures outside air temperature (OAT) to compute true air speed
- interfaced with IADS - provides IAS, TAS, Baro altitude, Total air temperature, Static temperature (OAT), Vertical speed and static pressure to IADS

AHS - attitude heading reference system

- These provides Helicopter attitudes, attitude rates, yaw rates, heading & accelerations to AFDS & to IAPS.
- magnetometer - provide heading information
- AHS has baro altitude, indicated air speed & True air speed.
- Vertical gyro: provides pitch & roll attributes to the AFCS.
- Radio altimeter: provides radio altitude & radio altitude validity to AFCS & IAPS.
- Doppler velocity sensor: provides longitudinal & lateral velocities to the AFCS.

Communication module

- ACU 500 LH communication system provide communication from air to air & air to ground in VHF and UHF bands.
- Also provide in ECCM mode (direct sequence spread spectrum) & SSF, frequency hopping, secure mode b/w similar or compatible users.

main Receiver covering range 30 to 406.475 MHz

Receive on of 4 frequencies: ~~40.5~~ ^(AM) ~~121.5~~ ~~156.8~~ ~~243~~
 MHz (AM) MHz (FM)

40.5 MHz (FM), 121.5 MHz (AM), 156.8 MHz (FM),
 243 MHz (AM)

MADPS (VDR System)

- MADPS: modular air borne data recording & acquisition system
- Crash survivable recording device, capable of acquiring and recording essential flight data & also voice data
- acquires from various on-board sensor / systems and records from at the rate of 256 up/s
- Retains upto 25 hours of flight data & 2 hours of voice data for 4 channels

Landing gear

- fixed, tri cycle, wheel type configuration with crashworthy features
- foot operated hydraulic wheel braking, parking brake available for pilots
- free casting, self centering tail wheel
- Capability of parking on lateral / longitudinal ground slope of 3° deg
- max forward velocity of landing - 100 kmph
- lashing points
- oleo pneumatic shock absorbers with 2 stage separated by ruptured disk
- Two stage levered suspension type shock absorbers
- Brake operations up to 65 kmph (35.1 knots) speed

Hard landing capability: 6.1 m/s with $L = 3/3$ w
 $L = \text{lift}$ $w = \text{Auw}$

Skis:

provision is available on landing gear to optionally install skis for operation on snow surface

Steering System:

- during taxiing is achieved by differential braking or tail rotor thrust.

tail wheel lock:

Tail landing gear is fitted with a pilot operated tail wheel lock through a control cable.